

KARAMAY, China

WMO: 512430

Lat: 45.6101N

Lon: 84.8454E

Elev: 1499

StdP: 13.92

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-9.6	-20.2	1.9	-13.6	-15.0	2.6	-8.5	20.6	18.2	10.4	14.6	1.8	340	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	97.9	64.8	95.2	64.0	92.7	63.4	67.5	88.0	66.3	87.9	65.3	87.1	7.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	85.1	75.1	58.6	77.8	75.8	56.4	71.7	76.3	32.8	87.6	31.8	88.0	31.0	87.1	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	20.5	16.1	-14.9	102.9	5.1	3.2	-18.6	105.3	-21.6	107.2	-24.5	109.0	-28.2	111.3		
(5)			-15.4	69.8	5.0	1.8	-19.0	71.0	-21.9	72.1	-24.8	73.1	-28.4	74.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	3.5	12.5	35.3	57.1	68.9	79.0	81.9	79.0	67.7	51.7	32.9	12.2
	DBStd	28.64	9.06	9.47	12.82	9.34	8.02	6.64	5.87	6.37	8.16	8.70	10.37	9.40
	HDD50	4789	1442	1049	479	42	1	0	0	0	2	87	514	1173
	HDD65	7692	1907	1469	920	261	53	1	0	1	63	417	962	1638
	CDD50	4302	0	0	24	256	588	869	990	900	533	140	2	0
	CDD65	1730	0	0	0	25	175	420	525	436	144	5	0	0
	CDH74	18730	0	0	0	164	1498	4692	6353	4882	1125	16	0	0
	CDH80	8555	0	0	0	32	515	2189	3195	2266	357	1	0	0
Wind	WSAvg	5.9	2.3	3.0	5.6	8.3	8.6	8.2	7.4	7.2	6.7	6.1	4.4	2.7
Precipitation	PrecAvg	4.1	0.1	0.1	0.2	0.3	0.6	0.6	0.9	0.6	0.3	0.2	0.2	0.2
	PrecMax	8.9	0.5	0.4	0.6	0.9	2.0	2.2	2.8	1.6	1.0	0.7	0.9	0.7
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.5	0.1	0.1	0.2	0.2	0.5	0.6	0.6	0.4	0.2	0.2	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	36.1	68.4	84.4	94.2	99.4	102.5	100.4	93.3	76.3	57.6	34.6
		MCWB	25.2	30.0	49.4	56.2	60.9	64.6	66.9	65.4	61.5	54.4	46.0	29.9
	2%	DB	22.5	31.8	62.0	78.3	89.3	96.3	98.6	96.5	87.8	71.1	52.3	31.1
		MCWB	20.5	27.6	47.1	53.9	59.5	63.4	65.8	64.5	59.9	51.9	42.8	28.1
	5%	DB	18.7	28.3	58.0	74.6	85.6	93.4	95.6	93.6	84.0	67.5	48.8	27.9
		MCWB	17.1	25.0	45.3	52.1	58.3	62.8	64.9	63.8	58.4	50.6	40.5	25.6
	10%	DB	15.8	25.2	53.8	70.9	82.1	90.6	92.8	90.5	80.5	64.1	45.8	24.9
		MCWB	14.5	22.8	43.1	50.9	57.0	62.4	64.4	63.1	56.8	49.1	38.4	22.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	31.3	51.1	58.3	62.9	67.9	70.1	69.1	62.9	56.5	47.3	31.5
		MCDB	27.7	34.7	65.7	77.0	87.1	87.2	94.9	85.8	85.7	71.3	55.1	33.2
	2%	WB	20.6	27.9	48.0	56.0	61.2	66.2	68.0	66.9	61.3	53.9	43.9	28.3
		MCDB	22.3	31.2	60.5	72.7	83.4	88.6	88.3	88.3	82.7	68.0	51.1	30.5
	5%	WB	17.2	25.2	45.7	54.2	59.8	64.9	66.8	65.5	59.9	51.8	41.1	25.8
		MCDB	18.5	28.1	57.3	71.1	81.0	87.4	87.7	87.7	80.2	65.1	48.5	27.9
	10%	WB	14.5	22.9	43.2	52.3	58.4	63.8	65.8	64.4	58.3	49.9	38.6	22.8
		MCDB	15.8	25.2	53.3	68.7	78.8	86.2	87.1	86.7	77.8	63.1	45.1	25.0
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.9	13.3	16.3	16.8	16.3	16.1	16.8	15.8	13.6	9.8	8.5
		MCDBR	10.3	11.6	17.0	19.4	19.5	18.3	18.1	19.2	18.8	16.9	12.8	8.8
		MCWBR	8.8	8.8	9.1	8.7	7.8	6.6	6.4	6.9	8.2	8.6	8.2	6.6
	5% WB	MCDBR	9.9	10.9	16.2	16.8	17.3	16.3	16.2	16.9	17.0	15.2	12.3	8.1
		MCWBR	8.5	8.4	9.0	8.3	7.3	6.0	6.0	6.3	8.0	8.1	8.1	6.2
Clear-Sky Solar Irradiance	taub	0.366	0.397	0.443	0.483	0.458	0.466	0.461	0.439	0.419	0.414	0.387	0.363	
	taud	1.994	1.959	1.932	1.901	2.022	2.069	2.125	2.159	2.139	2.058	2.058	2.016	
	Ebn at Noon	215	234	243	246	258	256	256	257	250	227	207	200	
	Edh at Noon	35	43	51	58	53	51	48	44	42	39	32	30	
All-Sky Solar Radiation	RadAvg	568	872	1289	1718	2089	2137	2076	1887	1518	987	584	430	
	RadStd	33	39	108	91	96	89	115	77	75	68	54	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	-0.83	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page